

Siddhant Thalal

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EDUCATION

IIT JODHPUR

B.TECH IN MECHANICAL ENGINEERING
SPECIALIZATION IN ROBOTICS

2021 - 2025

CGPA : 8.01

GONDIA PUBLIC SCHOOL

HIGH SCHOOL

2017 - 2019

Percentage : 89.4

GONDIA PUBLIC SCHOOL

INTERMEDIATE

2019 - 2021

Percentage : 87.6

SKILLS

ROBOTICS AND AUTOMATION

- Iot • Ros2 • Matlab/Simulink • Control systems
- Embedded Systems • Sensor integration • CAD

PROGRAMMING LANGUAGES

- Java • C

Familiar:

- Matlab • Python • PLC programming (Ladder Logic)

VCS & BUILD TOOLS

- Git • Github

FRONTEND

- HTML • CSS • Java Script • Bootstrap

BACKEND

- Node.js • Express

DATABASE MANAGEMENT

- MongoDB • MySQL

AWARDS/ACHIEVEMENTS

International Mathematics Olympiad Zonal Rank 17.

Secured an A grade in Data Structures and Algorithms course

COURSEWORK

- Mechatronics
- Digital fabrication
- Autonomous Systems
- Introduction to Robotics - Experimental Robotics
- Control of mechanical system
- Introduction to Cyber-Physical Systems
- Mobile Robots

PROJECTS

LAND ROVER, JAGUAR (INTER IIT- TECHMEET) ROBOTIC CHARGING CHALLENGE | CAD, OPEN CV, EMBEDDED SYSTEM, CONTROL

- An ArUco-based system was created to locate the charging socket using marker detection, and then, based on the input coordinates, the robot autonomously plugged the socket. It involved the development of an intelligent system combining computer vision and robotics. Tasks included robot ideation and movement planning, ArUco marker-based port detection, angle estimation using vector analysis, and trajectory generation via inverse kinematics.

CAMEL-INSPIRED BIOMIMETIC LEG FOR LOCOMOTION ON SANDY TERRAIN | ARDUINO, CAD, SIMSCAPE, FSR, ARUCO, CONTROL, IOTS

- Developed a camel foot-inspired leg that demonstrated significantly improved stability and reduced penetration on sand compared to conventional designs. Achieved smoother trajectory tracking using inverse kinematics and validated performance through dynamic simulations (SimScape) and real-world tests with ArUco-based motion tracking and FSR-based force analysis.

ROBOGUIDE: AUTONOMOUS CAMPUS NAVIGATOR FOR COLLEGE STUDENTS | PATH PLANNING, EMBEDDED SYSTEM, CONTROL, IOTS

- Built an autonomous navigation Robot that successfully executed GPS-based path planning and real-time obstacle avoidance using LIDAR and ultrasonic sensors. Enabled smooth navigation through dynamic environments by integrating motor control algorithms and sensor fusion. Achieved reliable end-to-end performance through full hardware-software integration and mobile app-based control.

TURTLEBOT3 & OPENMANIPULATOR-X | ROS2, GAZEBO, RVIZ

- Performed SLAM-based navigation of TurtleBot3 from point to point using ROS2 and Python, in both Gazebo and real-world environments. Modeled OpenMANIPULATOR-X with DH parameters and implemented inverse kinematics and motion control for grasping tasks through custom code and various manipulation experiments.

PORTFOLIO WEBSITE | HTML, CSS, JAVASCRIPT

- Built a responsive site to showcase robotics projects with interactive JavaScript features and smooth UI.
– [Portfolio Link](#)